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0610/62

October/November 2019

1 hour

No Additional Materials are required.

READ THESE INSTRUCTIONS FIRST

DO **NOT** WRITE IN ANY BARCODES.

Answer **all** questions.

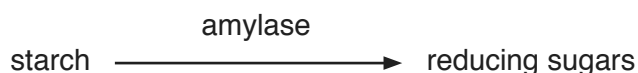
You may lose marks if you do not show your working or if you do not use appropriate units.

The number of marks is given in brackets [] at the end of each question or part question.

This syllabus is regulated for use in England, Wales and Northern Ireland as a Cambridge International Level 1/Level 2 Certificate.

This document consists of **11** printed pages and **1** blank page.

- 1 A student investigated the effect of the enzyme amylase on the breakdown of a starch suspension. The starch suspension was placed inside a bag made from dialysis tubing.



Dialysis tubing is made from a type of membrane that is partially permeable. Only small molecules can pass through this membrane.

- Step 1 Two large test-tubes were put into a water-bath. Each large test-tube contained 20 cm³ of distilled water. The temperature of the water-bath was 40 °C.
- Step 2 A knot was tied at the end of one piece of dialysis tubing, to form a bag.
- Step 3 5 cm³ of the starch suspension was put into the dialysis tubing bag.
- Step 4 A clean syringe was used to put 5 cm³ of amylase solution into the dialysis tubing bag.
- Step 5 The contents of the dialysis tubing bag were mixed well and the outside of the bag was rinsed with distilled water.
- Step 6 The student repeated steps 2 to 5 using 5 cm³ of distilled water in step 4 instead of the amylase solution.
- Step 7 The dialysis tubing bags were placed into the large test-tubes as shown in Fig. 1.1.

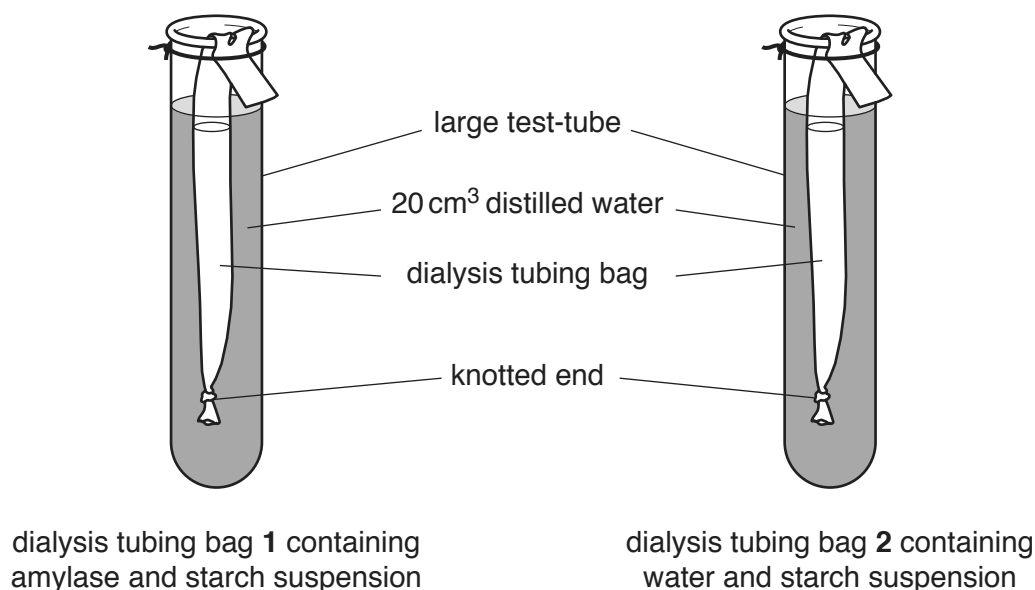


Fig. 1.1

- Step 8 Both large test-tubes were placed back into the water-bath and left for 10 minutes.
- Step 9 After 10 minutes the dialysis tubing bags were removed from the two large test-tubes.
- Step 10 The contents of the two large test-tubes and the two dialysis tubing bags were tested for reducing sugars and starch.

- (a) State the name of the solution that the student would use to test substances for starch.

..... [1]

- (b) Describe how the student would test substances for reducing sugars.

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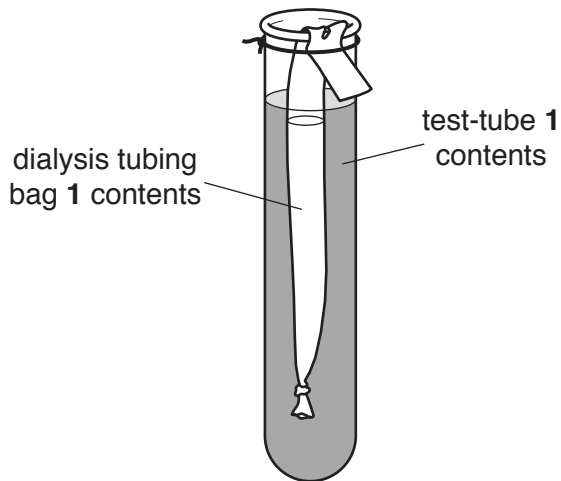
..... [2]

- (c) Suggest why the outside of the dialysis tubing bag was rinsed in Step 5.

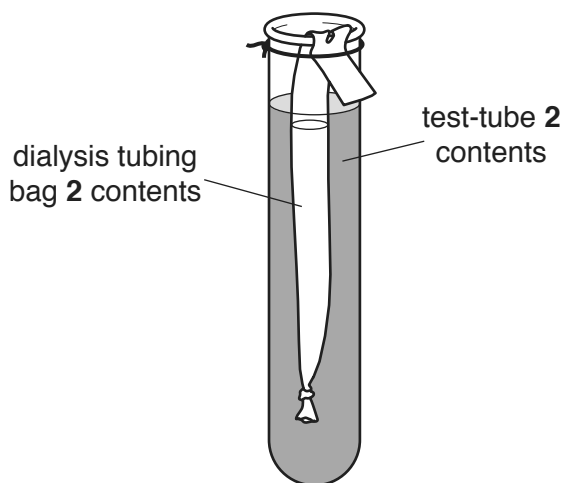
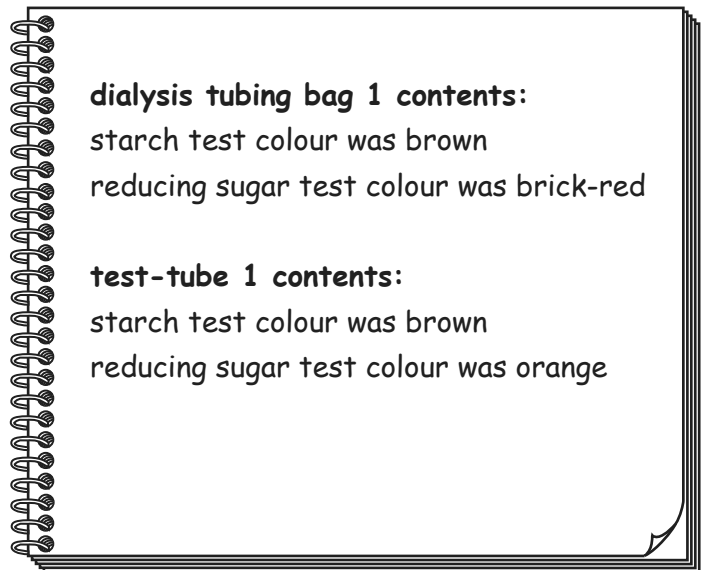
.....

..... [1]

The student's observations for the starch and reducing sugar tests in step 10 are shown in Fig. 1.2.



dialysis tubing bag 1 containing amylase and starch suspension at the start of the investigation



dialysis tubing bag 2 containing water and starch suspension at the start of the investigation

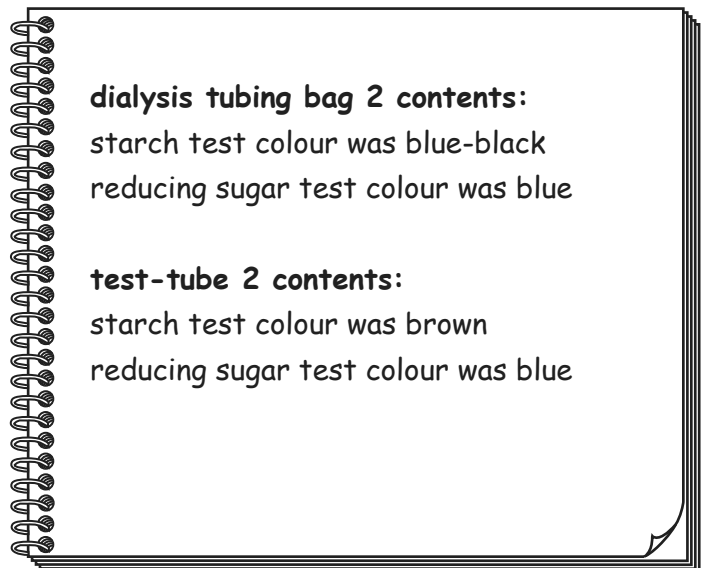


Fig. 1.2

- (d) (i) Prepare a table and record the results of the starch and reducing sugar tests from Fig. 1.2 in the space provided.

[3]

- (ii) State **three** conclusions for the results shown in Fig. 1.2.

.....

.....

.....

.....

.....

.....

..... [3]

- (e) State **two** variables that were kept constant during this investigation.

1

2

[2]

- (f) Plan an investigation **using dialysis tubing** to determine the effect of temperature on the activity of the enzyme amylase.

[6]

- (g) The student wanted to find out if the amylase enzyme passed through the dialysis tubing into the large test-tube. Amylase is made of protein.

Describe how the student could find out if the solution in the large test-tube contained protein.

..... [2]

[Total: 20]

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- 2 An investigation was performed to determine the effect of light intensity on leaf size in one species of plant.

Plants were grown in three different light intensities. The maximum width of each leaf was recorded. The results were recorded in Table 2.1 and an average value was calculated.

The results for three leaves grown in high light intensity are shown in Fig. 2.1. The horizontal line on each leaf indicates its maximum width.

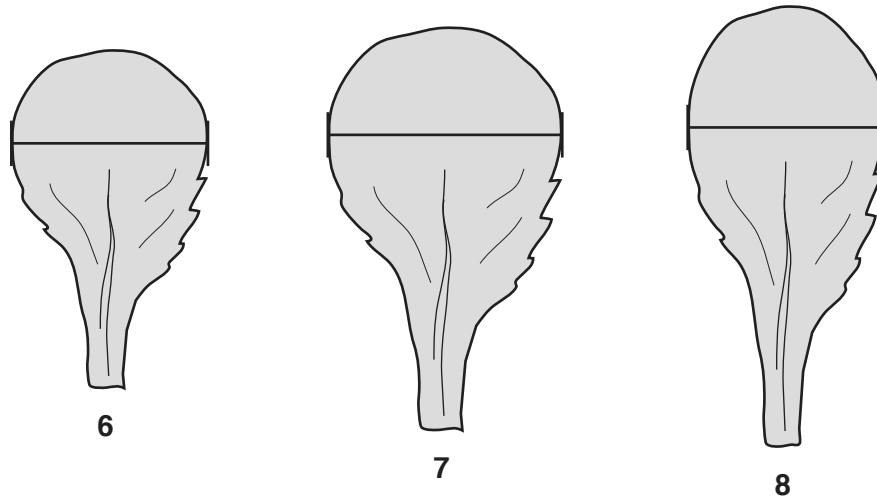


Fig. 2.1

- (a) (i) Measure the widths of leaves 6, 7 and 8 in Fig. 2.1 and record these values in Table 2.1. [1]

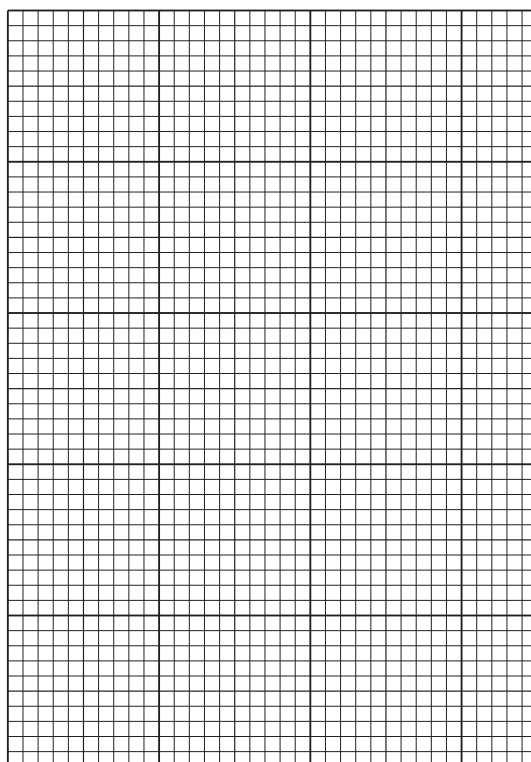
Table 2.1

leaf	maximum width of leaves / mm		
	low light intensity	medium light intensity	high light intensity
1	15	43	27
2	12	45	32
3	13	48	26
4	13	44	28
5	15	47	27
6	14	43	
7	12	12	
8	15	46	
average	14	41	

- (ii) Calculate the average width of the leaves grown in a high light intensity in Table 2.1. Record this value in Table 2.1.

[1]

- (iii) Plot a bar chart on the grid of the **average** leaf width for leaves grown in low, medium and high light intensity using the data in Table 2.1.



[3]

- (iv) Circle **one** measurement in Table 2.1 that could be considered to be anomalous.

Give a reason for your choice.

reason

..... [2]

- (b) (i) State the variable that was changed in this investigation (the independent variable).

..... [1]

- (ii) State the variable that was measured in this investigation (the dependent variable).

..... [1]

(c) Fig. 2.2 is a photomicrograph of a cross-section of a root.

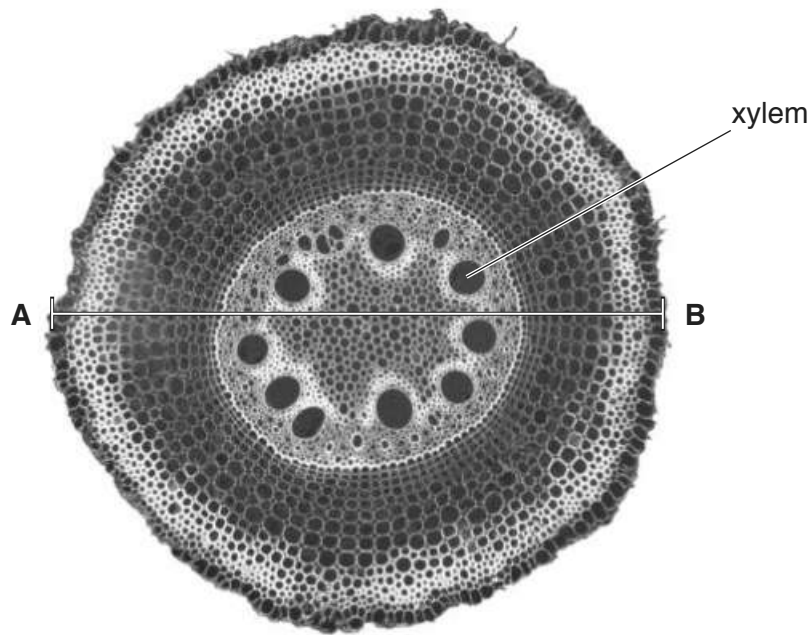


Fig. 2.2

- (i) Make a large drawing of the cross-section of the root in Fig. 2.2 to show the different areas of the root.

Do **not** draw individual cells.

- (ii) Measure line **AB** on Fig. 2.2 in millimetres.

length of line **AB** mm

The actual diameter of the root shown in Fig. 2.2 is 2 mm.

Calculate the magnification of Fig. 2.2 using the equation.

$$\text{magnification} = \frac{\text{length of line AB on Fig. 2.2}}{\text{actual diameter of the root}}$$

..... [2]

- (iii) Fig. 2.3 is a photomicrograph of a cross-section of a stem.

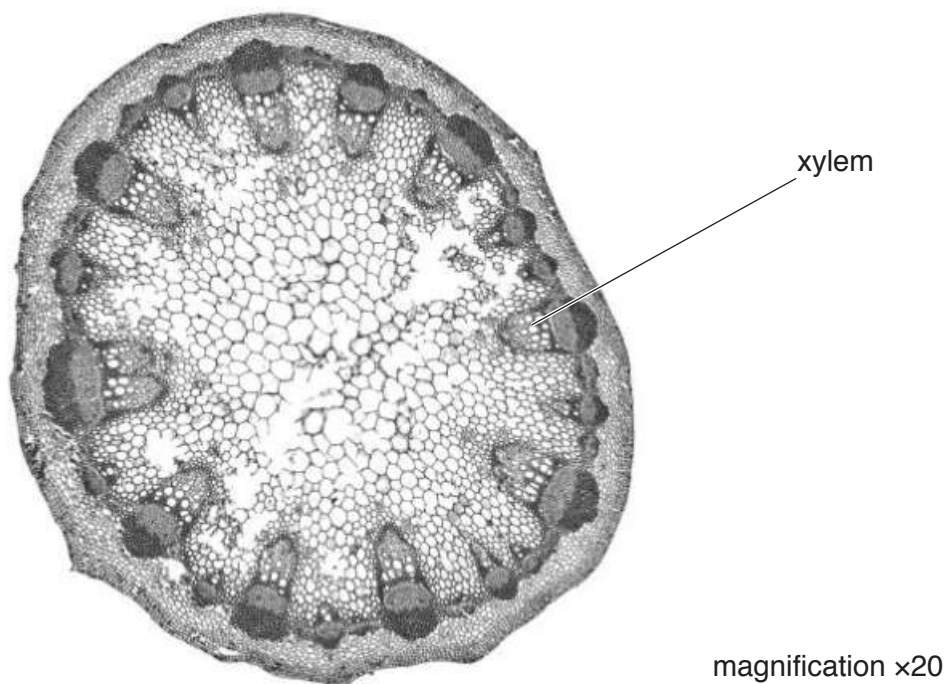


Fig. 2.3

State **two** differences between the root in Fig. 2.2 and the stem in Fig. 2.3.

- 1
-
- 2
-

[2]

(d) A student suggested that measuring leaf area is better than measuring leaf width.

(i) Describe how the area of a leaf could be measured.

.....

.....

.....

.....

..... [2]

(ii) Suggest why measuring leaf area is better than measuring leaf width.

.....

..... [1]

[Total: 20]

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